ModelIt!

Teacher's Guide

# Why Are Cheetahs Losing Their Speed?

Genetic Diversity, Habitat Loss, and the Future of the Fastest Animal

## Quick Reference

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|---|---|
| **Grade Level** | Grade 07L2 |
| **NGSS Standard** | MS-LS4-6, MS-LS4-4 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |

ModelIt! — Discovery Collective

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# 1 NGSS Standards & CAST Alignment

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**Performance Expectation: MS-LS4-6, MS-LS4-4**

Use mathematical representations to support explanations of how natural selection may lead to increases and decreases of specific traits in populations over time. Construct an explanation based on evidence that describes how genetic variations of traits in a population increase some individuals' probability of surviving and reproducing in a specific environment.

## Three-Dimensional Learning

| | |
|---|---|
| **Science Practice** | **Using Mathematics and Computational Thinking** — Students use mathematical representations in their model to show how population size and fragmentation drive changes in genetic diversity and species fitness over time. |
| **Disciplinary Core Idea** | **LS4.C Adaptation / LS4.B Natural Selection** — Natural selection acts on trait variation within populations. When genetic diversity is lost, the raw material for natural selection disappears, reducing a species' ability to adapt. |
| **Crosscutting Concept** | **Cause and Effect** — Students trace the causal chain from habitat fragmentation and population decline through reduced genetic diversity to decreased trait variation and species fitness. |

## CAST Testing Alignment

**MULTIPLE CHOICE:**

A wildlife biologist discovers that a population of 50 isolated cheetahs has much less genetic diversity than a connected population of 500 cheetahs. Which factor best explains this difference?

**CONSTRUCTED RESPONSE:**

Using your population genetics model, explain why cheetahs are more vulnerable to a new disease than a species like coyotes, which have high genetic diversity. Describe how genetic diversity affects disease resistance at the population level.

# 2 Scoring Rubric

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### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

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**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Cheetahs are a genetic paradox: the fastest land animal is also one of the most genetically uniform mammals on Earth. Their story reveals how population bottlenecks, habitat fragmentation, and the loss of genetic diversity can threaten even the most spectacular species.

## The Cheetah Bottleneck

About 10,000 years ago, during the last ice age, cheetah populations crashed to perhaps fewer than 500 individuals. This genetic bottleneck meant that all surviving cheetahs were closely related. Today, any two cheetahs are as genetically similar as identical twins in other species. Scientists discovered this when skin grafts between unrelated cheetahs were not rejected, something that normally only works between identical twins.

## Why Genetic Diversity Matters

Genetic diversity is a population's toolkit for survival. When individuals vary genetically, some will have traits that help them survive new diseases, climate changes, or predators. Without variation, the entire population is equally vulnerable. If a new disease kills one genetically identical cheetah, it can potentially kill them all.

## Habitat Fragmentation Compounds the Problem

Cheetahs once roamed across Africa, the Middle East, and into India. Today, they occupy only 9% of their historical range, broken into isolated patches by roads, farms, and cities. These isolated groups cannot breed with each other, so each small population's genetic diversity decreases further with each generation through genetic drift.

## Natural Selection Needs Variation

Natural selection is the engine of adaptation, but it needs fuel: genetic variation. If all cheetahs run at nearly the same speed (because they're nearly genetically identical), selection cannot favor faster individuals. The trait cannot improve. This is why the fastest animal on Earth may actually be getting slower over generations.

# 4 The LEVER Framework

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### About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L** **Locate the System**
Students identify population size, habitat fragmentation, genetic diversity, trait variation, and species fitness as components of the population genetics system.

**E** **Establish Relationships**
Students determine that population size and habitat fragmentation are external drivers that affect genetic diversity, which cascades to trait variation and species fitness.

**V** **Visualize & Model**
Students build a model showing how environmental pressures on population size and habitat connectivity determine the genetic health of a species.

**E** **Evaluate Outcomes**
Students run bottleneck, connected habitat, and recovery scenarios to observe how genetic diversity and species fitness respond to different population conditions.

**R** **Revise & Extend**
Students add disease susceptibility or gene flow to explore how genetic uniformity creates vulnerability and how migration can restore diversity.

### Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

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|---|---|
| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes. |
| **As a “what-if” tool** | When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

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## Slide 1: Cover

2 min

**Say:** *"Cheetahs can run 70 mph. They're the fastest animal on land. But scientists say they're getting slower. How is that even possible?"*

**Watch For**

Show slow-motion cheetah running footage. Ask: What would make the fastest animal on Earth slow down?

## Slide 2: Learning Objectives

3 min

**Say:** *"Today we're investigating a genetic mystery that threatens an entire species."*

**Watch For**

Have students read objectives. Pre-teach genetic diversity and habitat fragmentation.

## Slide 3: Big Question

3 min

**Say:** *"Here's the weird part: all cheetahs are almost genetically identical. You could swap skin between two random cheetahs and their bodies wouldn't reject it. Why is that a problem?"*

**Watch For**

Show genetic similarity comparison: cheetahs vs. other big cats. Discuss.

## Slide 4: LEVER Framework

2 min

**Say:** *"We'll build a model of how genetic diversity affects species survival."*

**Watch For**

Preview LEVER steps. Show cheetah historical vs. current range map.

## Slide 5: Activity 1: Components

8 min

**Say:** *"What factors affect a species' genetic health? Which are the causes and which are the effects?"*

**Watch For**

Guide sorting. Discuss why population size and fragmentation are the external drivers.

## Slide 6: Activity 2: Connections

8 min

**Say:** *"When population size drops, what happens to genetic diversity? And when genetic diversity drops, what happens to the species?"*

**Watch For**

Help students trace the causal chain. Emphasize that this is a cascade effect.

## Slide 7: Activity 3: Simulation

10 min

**Say:** *"Let's crash the population and see what happens to genetic diversity. Then let's try to bring it back."*

**Watch For**

Students predict first, then run all three scenarios. Discuss why recovery is so slow.

## Slide 8: Discoveries

5 min

**Say:** *"So can cheetahs get their speed back? What does our model say about their future?"*

**Watch For**

Lead discussion about genetic rescue and conservation genetics. Connect to natural selection.

## Slide 9: STEM Challenge

12 min

**Say:** *"You're conservation geneticists now. Design a plan to save the cheetah's genetic future."*

**Watch For**

Groups create conservation plans using model data and habitat maps.

# 6 Answer Key & Expected Responses

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## Component Sorting

- **EXTERNAL:** Population Size, Habitat Fragmentation
- **INTERNAL:** Genetic Diversity, Trait Variation, Species Fitness

## Relationships

- **Population Size to Genetic Diversity:** POSITIVE (+) — Larger populations maintain more genetic diversity because there are more individuals carrying different alleles. Small populations lose rare alleles through genetic drift.
- **Habitat Fragmentation to Genetic Diversity:** NEGATIVE (-) — Greater fragmentation isolates populations, preventing gene flow between groups. Each isolated group loses diversity independently through drift and inbreeding.
- **Genetic Diversity to Trait Variation:** POSITIVE (+) — More genetic diversity means more variation in physical traits like speed, disease resistance, and body size. Less diversity means individuals are more similar.
- **Trait Variation to Species Fitness:** POSITIVE (+) — Greater trait variation gives natural selection more to work with. Populations with diverse traits are better equipped to survive diseases, predators, and environmental changes.
- **Habitat Fragmentation to Population Size:** NEGATIVE (-) — Fragmented habitats support fewer individuals because each isolated patch has limited resources, increasing competition and reducing breeding opportunities.

## Key Discoveries

- Cheetahs have extremely low genetic diversity because they went through a population bottleneck about 10,000 years ago when their numbers dropped dramatically
- Low genetic diversity means less trait variation, so individuals are more similar and the population has fewer options for adapting to new threats
- Habitat fragmentation isolates small groups, preventing gene flow between populations and further reducing genetic diversity
- Natural selection can only work when there is genetic variation. Without variation, a population cannot adapt, no matter how strong the selection pressure

# 7 Collaborative Learning Protocols

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### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
|---|---|
| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
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| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

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### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

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### Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
|---|---|
| "Evolution always makes species better" | Evolution through natural selection improves adaptation to CURRENT conditions, but it can only work if there is genetic variation. Without variation, natural selection has nothing to select from, and the population cannot adapt. Cheetahs show that evolution can be stuck if genetic diversity is lost. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Endangered species just need more habitat" | Habitat is essential, but genetic diversity is equally important. A species with plenty of habitat but no genetic diversity is still vulnerable. Cheetahs could have all the habitat in the world, but without genetic variation, they cannot adapt to new diseases or environmental changes. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Small populations will naturally recover on their own" | Small populations actually lose genetic diversity faster than they can gain it. Mutations introduce new variation very slowly, while genetic drift removes it quickly in small groups. Without intervention (like wildlife corridors or genetic rescue), small populations spiral toward genetic collapse. |